Register No	RA2512044015071

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY
21CSC529T – Inferential Statistics
CYCLE TEST – I

Program Offered: M. Tech
Max. Marks: 40

Year / Semester: I / I
Duration: 1.5 Hrs
Date of Exam:

Part - A (2 * 20 = 40 Marks)

Answer any two Questions

Set - A

1.

a) A retail company seeks to compare the sales performance of two stores (Store 1 and Store 2). The sales figures are measured in thousands of dollars, and the data for 5 sales representatives from each store is provided below.

Data:

Store 1	120	130	150	140	160
Store 2	115	125	135	128	140

Question:

Using the given data:

i. Calculate the **mean, median,** and **mode** (if any) for both departments to understand the central tendency of productivity. (5 marks)

i. Compute the **range, variance,** and **standard deviation** for both departments to measure the dispersion. (7marks)

ii. Based on your analysis, which store shows higher consistency in sale Provide a statistical justification for your conclusion. (3 marks)

b. Explain the differences between correlation and covariance, and discuss their importance. (5 marks)

2.

a. For the given data (5 Marks)

	Football (F)	Basketball (B)	Tennis (T)
Adult	50	30	20
Teen	40	50	30
Child	60	40	50

i. What is the probability that a randomly chosen participant is enrolled in Basketball (B)?
ii. What is the probability that a randomly chosen participant is a Teen?
iii. What is the probability that a randomly chosen participant is an Adult enrolled in Football (F)?
iv. What is the probability that a randomly chosen participant is a Child enrolled in Tennis (T)?
v. What is the probability that a participant is an Adult given that the participant is enrolled in Football (F)?

b. A sports equipment manufacturer produces three types of items - footballs, basketballs, and baseballs. The probability of producing a defective football is 0.01, a defective basketball is 0.02, and a defective baseball is 0.03. The probability that a defective item is a football is 0.4, a basketball is 0.4, and a baseball is 0.2. If a defective item is picked up from the manufacturing line, what is the most likely type of item? (10 Marks)

c.Discuss in detail the measures of central tendency and provide an example to illustrate your explanation. (5 Marks)

3.

a. A health researcher is examining the effects of two factors, 'Sleep Duration' and 'Stress Level,' on individuals' 'Mental Well-Being.' The following data is collected for 5 individuals over a specific time period:

S.NO	Sleep Duration (hours)	Stress Level (on a scale of 1-10)	Mental Well-Being Score (out of 100)
1	7	3	80
2	5	7	65
3	8	2	90
4	6	5	75
5	4	9	60

Determine which independent variable (Sleep Duration or Stress Level) has a stronger relationship with Mental Well-Being. (10 marks)

b. Outline the different types of probabilistic sampling techniques and also provide examples for each (5 Marks)

c. A restaurant has a special menu featuring 5 unique dishes. The chef wants to create a tasting menu by selecting and arranging 3 out of these 5 dishes. How many different ways can the chef arrange the selected dishes on the tasting menu? (5 Marks)

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